

Midterm 2 (practice version - NOT GRADED), Math 170a, Fall 2018
Instructor: Elizaveta Rebrova

Printed name: _____

Signed name: _____

Student ID number: _____

Instructions:

- Read problems very carefully. If you have any questions please ask.
- The correct final answer alone is not sufficient for full credit - you should explain your answers as much as you can, saving enough time to attempt all the problems. Exception: Problem 1 (see instruction in the beginning of the statement).
- Your final answers do not need to be completely simplified (unless otherwise stated), e.g. a sum of several decimal fractions would be completely fine. Ask if in doubt.
- You are allowed to use only the items necessary for writing. Notes, books, sheets, calculators, phones are not allowed, please put them away. If you need more paper please raise your hand.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
Total:	50	

1. For this problem only: please provide short answers (or short explanations when appropriate).
 - (a) (2 points) Suppose that X is a random variable with the following PDF: $f(x) = cx^2$ when $1 \leq x \leq 2$ and 0 otherwise. Find c .
 - (b) (2 points) If random variables X and Y are independent express $\text{Var}(X - Y)$ in terms of $\text{Var}(X)$ and $\text{Var}(Y)$.
 - (c) (2 points) Let X be a random variable with expectation 0 and variance 1. Compute $\mathbb{E}(X^2 + X)$.
 - (d) (2 points) Two random variables X and Y both have range 0, 1, 2. Their joint PMF is given by $p_{X,Y}(x, y) = \frac{1}{3}y(x - 1)e^x$. Calculate $\mathbb{E}X$.
 - (e) (2 points) Prove that X and Y from previous question cannot be independent.

2. (10 points) Let X_1, X_2, Y_1, Y_2 be four independent identically distributed random variables. If D denotes the distance between the points (X_1, Y_1) and (X_2, Y_2) show that

$$\mathbb{E}(D^2) = 4 \operatorname{Var}(X_1).$$

3. (10 points) There are n couples coming to your dinner. These $2n$ people will be seated around a circular table which has exactly $2n$ seats in a uniformly random order. A couple will be happy if the partners are sitting next to each other or if there is exactly one person sitting between them.
- What is the probability that the first couple is happy?
 - Compute the expected number of happy couples.

4. (10 points) We are given a random number X of fair coins. Assume that X has Poisson distribution with parameter 2. We toss each of the coins independently, and let Y be the number of heads that come up. Show that Y has Poisson distribution with parameter 1.

5. (10 points) A factory has a machine that prints circuit boards. Each circuit board has a probability of p of jamming the machine. Once the machine jams, it destroys the circuit board that jammed it and stops working. Each circuit board contains a circuit with 6 components. In order to be working, at least 5 of these components must be functional. Each component is functional with probability a , independently.
- What is the expected number of circuit boards printed (working or not, not including the destroyed board)?
 - What is the expected number of working circuit boards? Simplify your answer (no summations should be in the final answer).